

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12411650
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Dundar; Bekir B. et al.

Generating a customized social-driven playlist

Abstract

Generating a customized playlist may include identifying a user account for which a playlist is to be generated, where the user account is associated with a user listening history, a taste profile, and a social profile, identifying one or more friend accounts linked to the user account based on the social profile, obtaining an indication of a plurality of songs associated with one or more of the friend accounts, determining a listening history for the one or more plurality of songs based on the friend accounts, obtaining a subset of the plurality of songs based on the listening history of the plurality of songs, and generating a playlist from the subset of the plurality of songs.

Inventors: Dundar; Bekir B. (Moraga, CA), Shenoy; Arvind S. (San Jose, CA), Cartoon; Daniel (South San Francisco, CA), Chen; Denise L. (Menlo Park, CA), Domm; Drew R. (Oakland, CA), Vinna; Fredric R. (San Francisco, CA), Levy; Mark H. (Winchester, GB), Irvine; Paul C. (Mill Valley, CA), Mustafi; Priyo (Fremont, CA), Alsina; Thomas (Saratoga, CA), Lindholm; Erik (Cupertino, CA), Schoberl; Uli M. (San Francisco, CA)

Applicant: Apple Inc. (Cupertino, CA)

Family ID: 69055268

Assignee: Apple Inc. (Cupertino, CA)

Appl. No.: 18/182667

Filed: March 13, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20230333808 A1	Oct. 19, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **G06F3/16** (20060101); **G06F3/0482** (20130101); **G06F16/2457** (20190101);
G06F16/638 (20190101); **H04L67/306** (20220101); **H04L67/50** (20220101)

U.S. Cl.:

CPC **G06F3/165** (20130101); **G06F3/0482** (20130101); **G06F16/24578** (20190101);
G06F16/639 (20190101); **H04L67/306** (20130101); **H04L67/535** (20220501);

Field of Classification Search

CPC: G06F (3/165); G06F (16/24578); G06F (16/639); G06F (3/0482); H04L (67/22); H04L
(67/306)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4607385	12/1985	Maeda	N/A	N/A
4996642	12/1990	Hey	N/A	N/A
5355302	12/1993	Martin	N/A	N/A
5375235	12/1993	Berry	N/A	N/A
5464946	12/1994	Lewis	N/A	N/A
5483278	12/1995	Strubbe	N/A	N/A
5583763	12/1995	Atcheson	N/A	N/A
5616876	12/1996	Cluts	N/A	N/A
5724521	12/1997	Dedrick	N/A	N/A
5754939	12/1997	Herz	N/A	N/A
5758257	12/1997	Herz	N/A	N/A
5765144	12/1997	Larche	N/A	N/A
5835087	12/1997	Herz	N/A	N/A
5890152	12/1998	Rapaport	N/A	N/A
5918014	12/1998	Robinson	N/A	N/A
5926624	12/1998	Katz	N/A	N/A
5931454	12/1998	Kondo	N/A	N/A
5950176	12/1998	Keiser	N/A	N/A
6000044	12/1998	Chrysos	N/A	N/A
6041311	12/1999	Chislenko	N/A	N/A
6047311	12/1999	Ueno	N/A	N/A
6112186	12/1999	Bergh	N/A	N/A
6134532	12/1999	Lazarus	N/A	N/A
6266649	12/2000	Linden et al.	N/A	N/A
6289353	12/2000	Hazlehurst et al.	N/A	N/A
6317722	12/2000	Jacobi et al.	N/A	N/A

6345288	12/2001	Reed	N/A	N/A
6346951	12/2001	Mastronardi	N/A	N/A
6347313	12/2001	Ma	N/A	N/A
6349339	12/2001	Williams	N/A	N/A
6381575	12/2001	Martin	N/A	N/A
6430539	12/2001	Lazarus	N/A	N/A
6434621	12/2001	Pezzillo	N/A	N/A
6438579	12/2001	Hosken	N/A	N/A
6487539	12/2001	Aggarwal	N/A	N/A
6526411	12/2002	Ward	N/A	N/A
6532469	12/2002	Feldman	N/A	N/A
6545209	12/2002	Flannery	N/A	N/A
6577716	12/2002	Minter	N/A	N/A
6587127	12/2002	Leeke	N/A	N/A
6615208	12/2002	Behrens	N/A	N/A
6647371	12/2002	Shinohara	N/A	N/A
6687696	12/2003	Hofmann	N/A	N/A
6690918	12/2003	Evans	N/A	N/A
6704576	12/2003	Brachman	N/A	N/A
6727914	12/2003	Gutta	N/A	N/A
6748395	12/2003	Picker	N/A	N/A
6751574	12/2003	Shinohara	N/A	N/A
6785688	12/2003	Abajian	N/A	N/A
6842748	12/2004	Warner	N/A	N/A
6842761	12/2004	Diamond	N/A	N/A
6850252	12/2004	Hoffberg	N/A	N/A
6865546	12/2004	Song	N/A	N/A
6914891	12/2004	Ha	N/A	N/A
6931454	12/2004	Deshpande	N/A	N/A
6933433	12/2004	Porteus	N/A	N/A
6941324	12/2004	Plastina	N/A	N/A
6947922	12/2004	Glance	N/A	N/A
6950804	12/2004	Strietzel	N/A	N/A
6987221	12/2005	Platt	N/A	N/A
6990497	12/2005	Orouke	N/A	N/A
6993532	12/2005	Platt	N/A	N/A
7013238	12/2005	Weare	N/A	N/A
7020637	12/2005	Bratton	N/A	N/A
7021836	12/2005	Anderson	N/A	N/A
7024424	12/2005	Platt	N/A	N/A
7043479	12/2005	Ireton	N/A	N/A
7051352	12/2005	Schaffer	N/A	N/A
7072846	12/2005	Robinson	N/A	N/A
7082407	12/2005	Bezos	N/A	N/A
7096234	12/2005	Plastina	N/A	N/A
7111240	12/2005	Crow	N/A	N/A
7113917	12/2005	Jacobi	N/A	N/A
7113999	12/2005	Pestoni	N/A	N/A
7117516	12/2005	Khoo	N/A	N/A
7120619	12/2005	Drucker	N/A	N/A

7127143	12/2005	Elkinsii	N/A	N/A
7133924	12/2005	Rosenberg	N/A	N/A
7136866	12/2005	Springer, Jr.	N/A	N/A
7139723	12/2005	Conkwright	N/A	N/A
7159000	12/2006	Plastina	N/A	N/A
7174126	12/2006	McElhatten	N/A	N/A
7180473	12/2006	Horie	N/A	N/A
7194421	12/2006	Conkwright	N/A	N/A
7196258	12/2006	Platt	N/A	N/A
7197472	12/2006	Conkwright	N/A	N/A
7224282	12/2006	Terauchi	N/A	N/A
7228054	12/2006	Cowgill	N/A	N/A
7236941	12/2006	Conkwright	N/A	N/A
7246041	12/2006	Fukuda	N/A	N/A
7256341	12/2006	Plastina	N/A	N/A
7277870	12/2006	Mourad	N/A	N/A
7296031	12/2006	Platt	N/A	N/A
7302419	12/2006	Conkwright	N/A	N/A
7302468	12/2006	Wijeratne	N/A	N/A
7340455	12/2007	Platt	N/A	N/A
7345232	12/2007	Toivonen	N/A	N/A
7358434	12/2007	Plastina	N/A	N/A
7363314	12/2007	Picker	N/A	N/A
7392212	12/2007	Hancock	N/A	N/A
7403769	12/2007	Kopra	N/A	N/A
7415181	12/2007	Greenwood	N/A	N/A
7434247	12/2007	Dudkiewicz	N/A	N/A
7457852	12/2007	Oroureke	N/A	N/A
7457862	12/2007	Hepworth	N/A	N/A
7478323	12/2008	Dowdy	N/A	N/A
7487107	12/2008	Blanchard	N/A	N/A
7490775	12/2008	Biderman	N/A	N/A
7493572	12/2008	Card	N/A	N/A
7499630	12/2008	Koch	N/A	N/A
7505959	12/2008	Kaiser	N/A	N/A
7546254	12/2008	Bednarek	N/A	N/A
7568213	12/2008	Carhart	N/A	N/A
7571121	12/2008	Bezos	N/A	N/A
7571183	12/2008	Renshaw	N/A	N/A
7574422	12/2008	Guan	N/A	N/A
7574513	12/2008	Dunning	N/A	N/A
7580932	12/2008	Plastina	N/A	N/A
7585204	12/2008	Kobayashi	N/A	N/A
7599847	12/2008	Block	N/A	N/A
7599906	12/2008	Kashiwagi	N/A	N/A
7599950	12/2008	Walther	N/A	N/A
7644077	12/2009	Picker	N/A	N/A
7647613	12/2009	Drakoulis	N/A	N/A
7650570	12/2009	Torrens	N/A	N/A
7653761	12/2009	Juster	N/A	N/A

7657224	12/2009	Goldberg	N/A	N/A
7657493	12/2009	Meijer	N/A	N/A
7680849	12/2009	Heller	N/A	N/A
7680959	12/2009	Svendsen	N/A	N/A
7685204	12/2009	Rogers	N/A	N/A
7693887	12/2009	McLaughlin	N/A	N/A
7707603	12/2009	Abanami	N/A	N/A
7720871	12/2009	Rogers	N/A	N/A
7725494	12/2009	Rogers	N/A	N/A
7734569	12/2009	Martin	N/A	N/A
7743009	12/2009	Hangartner	N/A	N/A
7797321	12/2009	Martin	N/A	N/A
7797446	12/2009	Heller	N/A	N/A
7801758	12/2009	Gracie	N/A	N/A
7826444	12/2009	Irvin	N/A	N/A
7831199	12/2009	Ng	N/A	N/A
7840570	12/2009	Martin Cervera	N/A	N/A
7853712	12/2009	Amidon	N/A	N/A
7875788	12/2010	Benyamin	N/A	N/A
7889724	12/2010	Irvin	N/A	N/A
8260656	12/2011	Harbick	N/A	N/A
8626607	12/2013	Wood	N/A	N/A
8639706	12/2013	Bilinski	707/752	G06F 16/285
8655464	12/2013	Dollar, Jr.	N/A	N/A
9131018	12/2014	Robertson	N/A	N/A
9269107	12/2015	Boulter	N/A	N/A
9495447	12/2015	Phillips	N/A	N/A
9720564	12/2016	Vinna	N/A	N/A
9870360	12/2017	Bilinski	N/A	N/A
10262060	12/2018	Zeiler	N/A	N/A
10853826	12/2019	Romagnolo	N/A	G06Q 30/0203
2001/0007099	12/2000	Rau	N/A	N/A
2001/0021914	12/2000	Jacobi	N/A	N/A
2001/0056434	12/2000	Kaplan	N/A	N/A
2002/0002899	12/2001	Gjerdingen	N/A	N/A
2002/0042912	12/2001	Iijima	N/A	N/A
2002/0045960	12/2001	Phillips	N/A	N/A
2002/0059094	12/2001	Hosea	N/A	N/A
2002/0082901	12/2001	Dunning	N/A	N/A
2002/0087970	12/2001	Dorricott	N/A	N/A
2002/0152117	12/2001	Cristofalo	N/A	N/A
2002/0178223	12/2001	Bushkin	N/A	N/A
2002/0178276	12/2001	McCartney	N/A	N/A
2002/0194215	12/2001	Cantrell	N/A	N/A
2003/0018709	12/2002	Schrempp	N/A	N/A
2003/0022953	12/2002	Zampini	N/A	N/A
2003/0033321	12/2002	Schrempp	N/A	N/A
2003/0055689	12/2002	Block	N/A	N/A
2003/0097379	12/2002	Ireton	N/A	N/A

2003/0120630	12/2002	Tunkelang	N/A	N/A
2003/0135513	12/2002	Quinn	N/A	N/A
2003/0212710	12/2002	Guy	N/A	N/A
2003/0221541	12/2002	Platt	N/A	N/A
2003/0229537	12/2002	Dunning	N/A	N/A
2004/0002993	12/2003	Toussaint	N/A	N/A
2004/0003392	12/2003	Trajkovic	N/A	N/A
2004/0055445	12/2003	Iyoku	N/A	N/A
2004/0068552	12/2003	Kotz	N/A	N/A
2004/0070538	12/2003	Horie	N/A	N/A
2004/0073924	12/2003	Pendakur	N/A	N/A
2004/0128286	12/2003	Yasushi	N/A	N/A
2004/0139064	12/2003	Chevallier	N/A	N/A
2004/0148424	12/2003	Berkson	N/A	N/A
2004/0158860	12/2003	Crow	N/A	N/A
2004/0162738	12/2003	Sanders	N/A	N/A
2004/0194128	12/2003	McIntyre	N/A	N/A
2004/0215657	12/2003	Drucker	N/A	N/A
2004/0252604	12/2003	Johnson	N/A	N/A
2004/0254659	12/2003	Bolas	N/A	N/A
2004/0263337	12/2003	Terauchi	N/A	N/A
2004/0267715	12/2003	Polson	N/A	N/A
2005/0004941	12/2004	Maria Kalker	N/A	N/A
2005/0019114	12/2004	Sung	N/A	N/A
2005/0021470	12/2004	Martin	N/A	N/A
2005/0038819	12/2004	Hicken	N/A	N/A
2005/0050079	12/2004	Plastina	N/A	N/A
2005/0060350	12/2004	Baum	N/A	N/A
2005/0075908	12/2004	Stevens	N/A	N/A
2005/0091146	12/2004	Levinson	N/A	N/A
2005/0091147	12/2004	Ingargiola	N/A	N/A
2005/0098023	12/2004	Toivonen	N/A	N/A
2005/0102610	12/2004	Jie	N/A	N/A
2005/0114357	12/2004	Chengalvarayan	N/A	N/A
2005/0131752	12/2004	Gracie	N/A	N/A
2005/0141709	12/2004	Bratton	N/A	N/A
2005/0154608	12/2004	Paulson	N/A	N/A
2005/0160458	12/2004	Baumgartner	N/A	N/A
2005/0193014	12/2004	Prince	N/A	N/A
2005/0193054	12/2004	Wilson	N/A	N/A
2005/0195696	12/2004	Rekimoto	N/A	N/A
2005/0198075	12/2004	Plastina	N/A	N/A
2005/0203807	12/2004	Bezos	N/A	N/A
2005/0210009	12/2004	Tran	N/A	N/A
2005/0210101	12/2004	Janik	N/A	N/A
2005/0210507	12/2004	Hawkins	N/A	N/A
2005/0216855	12/2004	Kopra	N/A	N/A
2005/0216859	12/2004	Paek	N/A	N/A
2005/0222989	12/2004	Haveliwala	N/A	N/A
2005/0223039	12/2004	Kim	N/A	N/A

2005/0234891	12/2004	Walther	N/A	N/A
2005/0235811	12/2004	Dukane	N/A	N/A
2005/0240494	12/2004	Cue	N/A	N/A
2005/0251440	12/2004	Bednarek	N/A	N/A
2005/0256867	12/2004	Walther	N/A	N/A
2005/0276570	12/2004	Reed	N/A	N/A
2005/0278377	12/2004	Mirrashidi	N/A	N/A
2006/0015571	12/2005	Fukuda	N/A	N/A
2006/0015904	12/2005	Marcus	N/A	N/A
2006/0018208	12/2005	Nathan	N/A	N/A
2006/0018209	12/2005	Drakoulis	N/A	N/A
2006/0020062	12/2005	Bloom	N/A	N/A
2006/0020662	12/2005	Robinson	N/A	N/A
2006/0026263	12/2005	Raghavan	N/A	N/A
2006/0032363	12/2005	Platt	N/A	N/A
2006/0053077	12/2005	Mourad	N/A	N/A
2006/0059225	12/2005	Stonehocker	N/A	N/A
2006/0062094	12/2005	Nathan	N/A	N/A
2006/0067296	12/2005	Bershad	N/A	N/A
2006/0074750	12/2005	Clark	N/A	N/A
2006/0080251	12/2005	Fried	N/A	N/A
2006/0080356	12/2005	Burges	N/A	N/A
2006/0091203	12/2005	Bakker	N/A	N/A
2006/0095516	12/2005	Wijeratne	N/A	N/A
2006/0100978	12/2005	Heller	N/A	N/A
2006/0107823	12/2005	Platt	N/A	N/A
2006/0112098	12/2005	Renshaw	N/A	N/A
2006/0123052	12/2005	Robbin	N/A	N/A
2006/0136344	12/2005	Jones	N/A	N/A
2006/0143236	12/2005	Wu	N/A	N/A
2006/0153040	12/2005	Girish	N/A	N/A
2006/0155754	12/2005	Lubin	N/A	N/A
2006/0156239	12/2005	Jobs	N/A	N/A
2006/0165571	12/2005	Seon	N/A	N/A
2006/0168340	12/2005	Heller	N/A	N/A
2006/0168616	12/2005	Candelore	N/A	N/A
2006/0171682	12/2005	Komano	N/A	N/A
2006/0173910	12/2005	McLaughlin	N/A	N/A
2006/0173916	12/2005	Sibley	N/A	N/A
2006/0174008	12/2005	Abanami	N/A	N/A
2006/0195462	12/2005	Rogers	707/999.005	G06Q 20/1235
2006/0195512	12/2005	Rogers	N/A	N/A
2006/0195513	12/2005	Rogers	N/A	N/A
2006/0195514	12/2005	Rogers	N/A	N/A
2006/0195515	12/2005	Beaupre	N/A	N/A
2006/0195516	12/2005	Beaupre	N/A	N/A
2006/0195521	12/2005	New	N/A	N/A
2006/0195789	12/2005	Rogers	N/A	N/A
2006/0195790	12/2005	Beaupre	N/A	N/A

2006/0204220	12/2005	Lee	N/A	N/A
2006/0206478	12/2005	Glaser	N/A	N/A
2006/0253874	12/2005	Stark	N/A	N/A
2006/0265421	12/2005	Ranasinghe	N/A	N/A
2006/0277098	12/2005	Chung	N/A	N/A
2006/0282304	12/2005	Bedard	N/A	N/A
2006/0282311	12/2005	Jiang	N/A	N/A
2006/0288367	12/2005	Swix	N/A	N/A
2007/0014536	12/2006	Hellman	N/A	N/A
2007/0016507	12/2006	Tzara	N/A	N/A
2007/0043829	12/2006	Dua	N/A	N/A
2007/0073596	12/2006	Alexander	N/A	N/A
2007/0100690	12/2006	Hopkins	N/A	N/A
2007/0101373	12/2006	Bodlanender	N/A	N/A
2007/0118546	12/2006	Acharya	N/A	N/A
2007/0136264	12/2006	Tran	N/A	N/A
2007/0156677	12/2006	Szabo	N/A	N/A
2007/0156732	12/2006	Surendran	N/A	N/A
2007/0161402	12/2006	Ng	N/A	N/A
2007/0162502	12/2006	Thomas	N/A	N/A
2007/0162546	12/2006	McLaughlin	N/A	N/A
2007/0165577	12/2006	Baker	N/A	N/A
2007/0174866	12/2006	Brown	N/A	N/A
2007/0183742	12/2006	Cowgill	N/A	N/A
2007/0203790	12/2006	Torrens	N/A	N/A
2007/0219996	12/2006	Jarvinen	N/A	N/A
2007/0220566	12/2006	Ahmad-Taylor	N/A	N/A
2007/0244880	12/2006	Martin	N/A	N/A
2007/0250429	12/2006	Walser	N/A	N/A
2007/0250761	12/2006	Bradley	N/A	N/A
2007/0265979	12/2006	Hangartner	N/A	N/A
2007/0271286	12/2006	Purang	N/A	N/A
2007/0271287	12/2006	Acharya	N/A	N/A
2007/0294096	12/2006	Randall	N/A	N/A
2008/0004948	12/2007	Flake	N/A	N/A
2008/0004990	12/2007	Flake	N/A	N/A
2008/0010266	12/2007	Brunn	N/A	N/A
2008/0021851	12/2007	Alcalde	N/A	N/A
2008/0027881	12/2007	Bisse	N/A	N/A
2008/0033979	12/2007	Vignoli	N/A	N/A
2008/0040326	12/2007	Chang	N/A	N/A
2008/0046317	12/2007	Christianson	N/A	N/A
2008/0065659	12/2007	Watanabe	N/A	N/A
2008/0077264	12/2007	Irvin	N/A	N/A
2008/0082467	12/2007	Meijer	N/A	N/A
2008/0091717	12/2007	Garbow	N/A	N/A
2008/0109378	12/2007	Papadimitriou	N/A	N/A
2008/0115173	12/2007	Ellis	N/A	N/A
2008/0120339	12/2007	Guan	N/A	N/A
2008/0126384	12/2007	Toms	N/A	N/A

2008/0133593	12/2007	Clark	N/A	N/A
2008/0133601	12/2007	Cervera	N/A	N/A
2008/0133737	12/2007	Fischer	N/A	N/A
2008/0147711	12/2007	Spiegelman	N/A	N/A
2008/0154942	12/2007	Tsai	N/A	N/A
2008/0155057	12/2007	Khedouri	N/A	N/A
2008/0155588	12/2007	Roberts	N/A	N/A
2008/0168059	12/2007	Hoashi	N/A	N/A
2008/0195438	12/2007	Manfredi	N/A	N/A
2008/0215173	12/2007	Hicken	N/A	N/A
2008/0220855	12/2007	Chen	N/A	N/A
2008/0250067	12/2007	Svendsen	N/A	N/A
2008/0256106	12/2007	Whitman	N/A	N/A
2008/0270221	12/2007	Clemens	N/A	N/A
2009/0006353	12/2008	Vignoli	N/A	N/A
2009/0024504	12/2008	Lerman	N/A	N/A
2009/0024510	12/2008	Chen	N/A	N/A
2009/0043811	12/2008	Yamamoto	N/A	N/A
2009/0048957	12/2008	Celano	N/A	N/A
2009/0055426	12/2008	Kalasapur	N/A	N/A
2009/0063414	12/2008	White	N/A	N/A
2009/0070267	12/2008	Hangartner	N/A	N/A
2009/0073174	12/2008	Berg	N/A	N/A
2009/0076939	12/2008	Berg	N/A	N/A
2009/0076974	12/2008	Berg	N/A	N/A
2009/0083307	12/2008	Cervera	N/A	N/A
2009/0089222	12/2008	Castro	N/A	N/A
2009/0106085	12/2008	Raimbeault	N/A	N/A
2009/0164641	12/2008	Rogers	N/A	N/A
2009/0210415	12/2008	Martin	N/A	N/A
2009/0276368	12/2008	Martin	N/A	N/A
2009/0276709	12/2008	Venneman	N/A	N/A
2010/0011388	12/2009	Bull	N/A	N/A
2010/0161595	12/2009	Martin	N/A	N/A
2010/0169328	12/2009	Hangartner	N/A	N/A
2010/0188405	12/2009	Haughay, Jr.	N/A	N/A
2011/0016120	12/2010	Haughay, Jr.	N/A	N/A
2011/0060738	12/2010	Gates	N/A	N/A
2011/0119127	12/2010	Hangartner	N/A	N/A
2011/0246508	12/2010	Maekawa	N/A	N/A
2011/0295843	12/2010	Ingrassia	N/A	N/A
2012/0030230	12/2011	Sheinkop	N/A	N/A
2012/0041967	12/2011	Askey	N/A	N/A
2012/0054666	12/2011	Baird-Smith	N/A	N/A
2012/0290648	12/2011	Sharkey	N/A	N/A
2013/0275353	12/2012	Ingrassia, Jr. et al.	N/A	N/A
2014/0025702	12/2013	Curtiss	707/769	G06Q 30/02
2014/0122477	12/2013	McFarlane	N/A	N/A
2014/0223099	12/2013	Kidron	N/A	N/A
2014/0230631	12/2013	Wieder	N/A	N/A

2014/0237361	12/2013	Martin	N/A	N/A
2014/0280181	12/2013	Rodger	N/A	N/A
2014/0282772	12/2013	Chen	N/A	N/A
2015/0033292	12/2014	Nguyen	N/A	N/A
2015/0039620	12/2014	Ning	N/A	N/A
2015/0039644	12/2014	Trivedi	N/A	N/A
2015/0089517	12/2014	Ruffini	725/9	H04N 21/4826
2015/0106320	12/2014	Boulter	N/A	N/A
2015/0153910	12/2014	Wheeler	N/A	N/A
2015/0193192	12/2014	Kidron	N/A	N/A
2015/0268800	12/2014	O'Konski	N/A	N/A
2015/0309768	12/2014	Van Der Heide	N/A	N/A
2015/0331940	12/2014	Manning	707/740	G06F 16/4387
2016/0066038	12/2015	Chesluk	N/A	N/A
2016/0162125	12/2015	Martella	N/A	N/A
2016/0196345	12/2015	Kreifeldt	N/A	N/A
2016/0292269	12/2015	O'Driscoll	N/A	N/A
2016/0299906	12/2015	Cartoon	N/A	N/A
2016/0335266	12/2015	Ogle	N/A	N/A
2016/0360271	12/2015	Magahern	N/A	N/A
2017/0092247	12/2016	Silverstein	N/A	N/A
2017/0161273	12/2016	Cao	N/A	N/A
2017/0161702	12/2016	Wood	N/A	N/A
2018/0015369	12/2017	Schupak	N/A	G06T 11/206
2018/0097932	12/2017	Cohen	N/A	H04M 7/0033
2018/0157745	12/2017	Williams	N/A	N/A
2018/0183739	12/2017	Foerster	N/A	N/A
2018/0189306	12/2017	Lamere	N/A	N/A
2018/0256078	12/2017	Vaterlaus	N/A	N/A
2018/0349011	12/2017	Morag	N/A	N/A
2019/0306105	12/2018	Snibbe	N/A	G10L 25/63
2019/0325035	12/2018	Sagui	N/A	G06F 16/4387

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
0643359	12/1994	EP	N/A
1050833	12/1999	EP	N/A
1231788	12/2001	EP	N/A
1420388	12/2003	EP	N/A
1548741	12/2004	EP	N/A
1906320	12/2007	EP	N/A
2161668	12/2009	EP	N/A
2306344	12/2010	EP	N/A
H1152965	12/1998	JP	N/A
2002108351	12/2001	JP	N/A

2002320203	12/2001	JP	N/A
2003255958	12/2002	JP	N/A
2004221999	12/2003	JP	N/A
2005027337	12/2004	JP	N/A
2008165759	12/2007	JP	N/A
2009048706	12/2008	JP	N/A
20020025579	12/2001	KR	N/A
0017793	12/1999	WO	N/A
2003019560	12/2002	WO	N/A
2003036541	12/2002	WO	N/A
2003051051	12/2002	WO	N/A
2004070538	12/2003	WO	N/A
2005013114	12/2004	WO	N/A
2005038666	12/2004	WO	N/A
2005046252	12/2004	WO	N/A
2005071569	12/2004	WO	N/A
2005106877	12/2004	WO	N/A
2005115107	12/2004	WO	N/A
2006040710	12/2005	WO	N/A
2006052837	12/2005	WO	N/A
2006075032	12/2005	WO	N/A
2006096664	12/2005	WO	N/A
2006097795	12/2005	WO	N/A
2006114451	12/2005	WO	N/A
2007038806	12/2006	WO	N/A
2007075622	12/2006	WO	N/A
2007092053	12/2006	WO	N/A
2007134193	12/2006	WO	N/A
2009149046	12/2008	WO	N/A

OTHER PUBLICATIONS

Pampalk, et al., "Content-based Organization and Visualization of Music Archives," ACM Multimedia, Juan les Pins, France, Dec. 2002, pp. 570-579. cited by applicant

Pauws, et al., "PATs: Realization and User Evaluation of an Automatic Playlist Generator," The International Conferences on Music Information Retrieval and Related Activities (ISMIR 2002), Paris, France, Oct. 2002, 9 pages. cited by applicant

Platt, et al., 'Learning a Gaussian Process Prior for Automatically Generating Music Playlists,' Microsoft Corporation {platt, cburgess, sswenson, chriswea}@microsoft.com, alcez@cs.berkeley.edu, pp. 1-9. cited by applicant

Platt, John C., 'Fast Embedding of Sparse Music Similarity Graphs,' Microsoft Corporation, {jplatt@microsoft.com}. cited by applicant

Porter, Dave, "Digital Music Sales Triple to \$1.1 Billion in 2005"

www.axcessnews.com/modules/wfsection/article.php?articleid=8327, Web Page, Feb. 24, 2006, Maintenance Fees. cited by applicant

Ragno et al., "Inferring Similarity Between Music Objects with Application to Playlist Generation," MIR'05, Nov. 10-11, 2005, Singapore, pp. 73-80. cited by applicant

Rauber, et al., "The SOM-enhanced JukeBox: Organization and Visualization of Music Collections Based on Perceptual Models," Journal of New Music Research, vol. 32, Nov. 2, 2003, pp. 193-210. cited by applicant

S. Baluja, R. Seth, D. Sivakumar, Y. Jing, J. Yagnik, S. Kumar, D. Ravichandran, and M. Aly,

“Video Suggestion and Discovery for YouTube: Taking Random Walks Through the View Graph,” In WWW '08: Proceedings of the 17th international conference on World Wide Web, pp. 895-904, Beijing, China, 2008. ACM Press. cited by applicant

Scheible, et al., “MobiLenin—Combining A Multi-Track Music Video, Personal Mobile Phones and a Public Display into Multi-User Interactive Entertainment,” Media Lab, University of Art and Design, Helsinki, Finland, pp. 1-10, Nov. 6-10, 2005. cited by applicant

Shneiderman, Ben, “Tree Visualization with Tree-Maps: 2-d Space-Filling Approach,” ACM Transactions on Graphics, vol. 11, No. 1, Jan. 1992, pp. 92-99. cited by applicant

Shneiderman, Ben, “Treemaps for Space-Constrained Visualization of Hierarchies,” <http://www.sc.umd.edu/theil/treemaphistory/>, last updated Apr. 28, 2006, 16 pages. cited by applicant

Smart Computing, ‘The Scoop on File-Sharing Services,’ Dec. 2000, vol. 11, Issue 12; pp. 30-33 in printed issue. Available at www.smartcomputing.com/editorial/article.asp?article=articles%2F2000%Fs11-12.degree./02F08s12.asp. cited by applicant

Steffen Pauws and Berry Eggen, “Realization and User Evaluation of an Automatic Playlist Generator,” Journal New Music Research, 2003, vol. 32, No. 2, pp. 179-192. cited by applicant

Strands Business Solutions. ‘Integration Document v.2.0,’ Published May 2008, [online retrieved on Jan. 21, 2010] Retrieved from the internet <URL: <http://recommender.strands.com/fdocfSBS-Integration-Document.pdf>>, entire document—18 pages. cited by applicant

T. Hofmann. ‘Latent Semantic Models for Collaborative Filtering’. ACM Transactions on Information Systems, 22:89-115, 2004. cited by applicant

T. Hofmann. ‘Unsupervised Learning by Probabilistic Latent Semantic Analysis’. Mach. Learn., 42:177-196, 2001. cited by applicant

Teng et al., “Design and Evaluation of mProducer: a Mobile Authoring Tool for Personal Experience Computing,” [online], MUM 2004, College Park, Maryland, USA, Oct. 27-29, 2004 [retrieved on Dec. 17, 2010]. [<http://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.131.2933&rep=rep1&type=pdf>]. cited by applicant

Tom Bunzel, ‘Easy Digital Music,’ QUE Publisher, Aug. 18, 2004, Chapters 5 and 8. cited by applicant

Treemap, University of Maryland, <http://www.cs.umd.edu/hcil/treemap/>, last updated Aug. 5, 2003, 4 pages. cited by applicant

Trustees of Indiana University, Variations2, The Indiana University Digital Music Library, <http://dmi.indiana.edu/>, last updated May 11, 2005. cited by applicant

Tzanetakis, et al., “MARSYAS3D: A Prototype Audio Browser-Editor Using a Large Scale Immersive Visual and Audio Display,” Proceedings of the 2001 International Conference on Auditory Display, Espoo, Finland, Jul./Aug. 2001, 5 pages. cited by applicant

Wolfers, Justin and Zitzewitz, Eric, “Prediction Markets,” Journal of Economic Perspectives, Spring 2004, pp. 107-126, vol. 18, No. 2. cited by applicant

www.akoo.com/Akoo/, Web Page, AKOO, Pick the Music, Waiting in the line at the Theme Park, Introducing the m-Venue.TM. platform. cited by applicant

www.ecastinc.com/music.sub.--licensing.html, Web Page, ECAST Network, interactive entertainment network, Music/Licensing. cited by applicant

www.roweinternational.com/jukeboxes.sub.--dia.html, Web Page, Digital Internet Access Jukeboxes, Rowe International. cited by applicant

www.touchtunes.com, Web Page, Touchtunes, Turn your ROWE 100A's and 100B's into touch tunes Digital Jukeboxes—Bose. cited by applicant

Y. Dempster, N. Laird, and D. Rubin. ‘Maximum Likelihood from Incomplete Data via the EM Algorithm’. Jour. of the Royal Stat. Soc., Ser. B., 39:1047-1053, 1977. cited by applicant

Yates et al., “ShopSmart: Product Recommendations through Technical Specifications and User Reviews,” Temple University, CIKM, Oct. 26-30, 2008, Napa Valley, CA, USA, 2 pages. cited by

applicant

Yen, Yi-Wyn, "Apple announces a 32GB iPhone 3G by Jun. 15, 2009?," The Industry Standard, Apr. 2, 2009, <http://www.thestandard.com/predictions/channel/hardware>, downloaded Apr. 8, 2009. cited by applicant

'Communications of the ACM' Dec. 1992, vol. 35, No. 12 at pp. 26-28 (Introduction to special issue regarding Workshop on High Performance Information Filtering, Morristown, N.J. Nov. 1991). cited by applicant

'New Music Recommendation System is Based on FOAF Personal Profiling,' www.masternewmedia.org/music_recommendation/music_recommendation_system/FOAF, Oct. 1, 2005. cited by applicant

'Social Networking Meets Music Listening: Mecora Launches Radio 2.0,' www.masternewmedia.org/news/2006/04/13/socialnetworking_meets_music_listening.htm, Apr. 13, 2006. cited by applicant

"Touch Tunes Signs License Agreement for BMI Music in Digital Jukeboxes," www.bmi.com/news/200403/20040324b.asp, Web Page, BMI .TM. Figures Don'T Lie, Mar. 24, 2004. cited by applicant

A. Das, M. Datar, A. Garg, and S. Rajaram. 'Google News Personalization: Scalable Online Collaborative Filtering'. In WWW '07: Proceedings of the 16th international conference on World Wide Web, pp. 271-280, New York, NY, USA, 2007. ACM Press. cited by applicant

Adcock et al., "AreWeThereYet?—A Temporally Aware Media Player," In Proc. 9th Australasian User Interface Conference (AUIC 2008), Wollongong, Aust. 2008, pp. 29-32. cited by applicant

Alvear, Jose, "Risk-Free Trial Streaming Media Delivery Tools," Streaming Media.com; www.streamingmedia.com/article.ap?id=5768, Jun. 30, 2000. cited by applicant

Andreja Andric and Goffredo Haus, "Estimating Quality of Playlists by Sight," AXMEDIS'05, IEEE Computer Society Washington, DC, 2005. cited by applicant

Apple: iTunes 4.2 User Guide for Windows; Dec. 2003; retrieved from the internet: URL: <http://www2.austin.cc.tx.us/tcm/projects/itunes.pdf>; pp. 10, 17-19. (Added Ref Nov. 5, 2009). cited by applicant

Arto Lehtiniemi and Jamo Seppanen, "Evaluation of Automatic Mobile Playlist Generator," IS-CHI 2007: Mobile devices and services, 2007, pp. 452-459. cited by applicant

Asa Rudstrom, "Co-Construction of Hybrid Spaces," A Dissertation submitted to the University of Stockholm in partial fulfillment of the requirements for the Degree of Doctor of Philosophy, Department of Computer and Systems Sciences Stockholm University and Royal Institute of Technology, pp. 1-69, Nov. 2005. cited by applicant

Belkin, et al., 'Information Filtering and Information Retrieval: Two Sides of the Same Coin?', Communications of the ACM (Dec. 1992). cited by applicant

Bender, 'Twenty Years of Personalization: All about the Daily Me,' Educause Review (Sep./Oct. 2002). cited by applicant

Bender, et al., 'Newspace: Mass Media and Personal Computing,' Proceedings of USENIX, Conference, pp. 329-348 (Summer 1991). cited by applicant

Brenner, Ryan, "Not Your Average Jukebox," On Hollywood 1000 contender Ecast users broadband to being the digital media experience to your watering hole, www.always-on-network.com/comments.php?id=P12663_0_37_0_C, Web Page. cited by applicant

Cano, Pedro et al., "On the Use of FastMap for Audio Retrieval and Browsing," The International Conference on Music Information Retrieval and Related Activities (ISMIR 2002), Paris, France, Oct. 2002, 2 pages. cited by applicant

Carlson et al. "Internet Banking: Market Developments and Regulatory Issues," In the New Economy: What Changed, and the Challenges for Economic Policy, May 2001, <http://www.occ.gov/netbank/SGEC2000.pdf>. cited by applicant

Connell, Lain et al., "Ontological Sketch Models: Highlighting User-System Misfits," in P.

Palanque, E. O'Neill and P. Johnson, editors, *Proceedings of Human Computer Interaction (HCI)* Bath, England, Sep. 2003, London Springer, pp. 1-16. cited by applicant

Crampes et al., "Automatic Playlist Composition in a Dynamic Music Landscape," *SADPI'07*, May 21-22, 2007, Montpellier, France, vol. 259, 2007, pp. 15-20. cited by applicant

Deshpande, Mukund, et al., 'Item-Based Top-N Recommendation Algorithms,' *ACM Transactions on Information Systems*, 22:1 (Jan. 2004), pp. 143-177. cited by applicant

Elias Pampalk, Arthur Flexer and Gerhard Widmer, "Improvements of Audio-Based Music Similarity and Genre Classification," in *Proceedings of 6th International Conference on Music Information Retrieval (ISMIR'05)*, pp. 628-633, London, UK, Sep. 2005. cited by applicant

Hijikata et al., "Content-based Music Filtering System with Editable User Profile," *Proc. of the 2006 ACM Symposium on Applied Computing*, Dijon, France, pp. 1050-1057. cited by applicant

I. Scihira., "A Characterization of Singular Graphs," *Electronic Journal of Linear Algebra*, 16:451-462, 2007. cited by applicant

Industry Standard, The, *Help FAQs for Standard Prediction Market*, <http://www.thestandard.com/help>, downloaded Jun. 29, 2009. cited by applicant

IP City, *Integrated Project on Interaction and Presence on Urban Environments—Demonstrators on Large-Scale Events Applications*; ipcity.eu; Giulio Jacucci, John Evans, Tommi Ilmonen; pp. 1-37; Feb. 9, 2007. cited by applicant

J. Dean and S. Ghemawat, 'MapReduce: Simplified Data Processing on Large Clusters'. *Commun. ACM*, 51 (1):107-113, 2008. cited by applicant

J.M. Buldu, P. Cano, M. Koppenberger, J. A. Almendral and S. Boccaletti, "The complex network of musical tastes," *New Journal of Physics*, vol. 9, No. 6, Jun. 2007, p. 172, IOP Publishing Ltd., Bristol, United Kingdom and Deutsche Physikalische Gesellschaft. cited by applicant

Jimeng Sun, Dacheng Tao, Spiros Papadimitriou, Philip Yu, Christos Faloutsos, "Incremental tensor analysis: theory and applications," *ACM*, Oct. 2008, pp. 1-37. cited by applicant

Johan Bollen, Herbert Van de Sompel, Joan Smith, Rick Luce, "Toward alternative metrics of journal impact: a comparison of download and citation data," *Google.com*, 2005, pp. 1-2. cited by applicant

John Thompson, 'A Graphic Representation of Interaction With the NEXIS News Database,' MIT Thesis (May 1983). cited by applicant

Jon Orwant, 'Apprising the User of User Models: Doppelganger's Interface,' in: A. Kobsa and D. Litman (eds.), *Proceeding of the 4th International Conference on User Modeling (1994)*. cited by applicant

Jonathan L. Orwant, 'Doppelganger Goes to School: Machine Learning for User Modeling,' MIT Master of Science Thesis (Sep. 1993). cited by applicant

Jonathan L. Orwant, 'Doppelganger: A User Modeling System,' MIT Bachelor's Thesis (Jun. 1991). cited by applicant

Leong et al., "Randomness as a Resource for Design," *Proceedings of the 6th conference on Designing Interactive systems*, 2006, University Park, PA, pp. 132-139. cited by applicant

Lie, 'The Electronic Broadsheet—All the News That Fits the Display,' MIT Master's Thesis, pp. 1-96 (Jun. 1991). cited by applicant

Lippman, et al., 'News and Movies in the 50 Megabit Living Room,' *IEEE/IEICE, Global Telecommunications Conference*, pp. 1976-1981 (Nov. 15, 1987). cited by applicant

Loeb et al., "Lessons from LyricTime™: A Prototype Multimedia System" *4th IEEE ComSoc International Workshop on Multimedia Communications (Apr. 1992)*. cited by applicant

Loeb, Shoshana, "Architecting Personalized Delivery of Multimedia Information," *Communications of the ACM* (Dec. 1992). cited by applicant

Loeb, Shoshana, "Delivering Interactive Multimedia Documents over Networks," *IEEE Communications Magazine*, May 1992, 8 pages. cited by applicant

Logan, Beth et al., "A Music Similarity Function Based on Signal Analysis," *IEEE International*

Conference on Multimedia and Expo (ICME), Tokyo, Japan, Aug. 2001, IEEE Press, pp. 952-955. cited by applicant

Logan, Beth, "Content-Based Playlist Generation: Exploratory Experiments," The International Conference on Music Information Retrieval and Related Activities (ISMIR 2002), Paris, France, Oct. 2002, 2 pages. cited by applicant

M. O'Connor, D. Cosley, J.A. Konstan, J. Riedl, "PolyLens: A Recommender System for Groups of Users," European Conference on Computer Supported Co-Operative Work at Bonn, Germany, Published 2001, pp. 199-218. cited by applicant

Maidin, Donncha O et al., "The Best of Two Worlds: Retrieving and Browsing," Proceedings of the Cost G-6 Conference on Digital Audio Effects (DAFX-00), Verona, Italy, Dec. 7-9, 2000, 4 pages. cited by applicant

McCarthy, et al., "Augmenting the Social Space of an Academic Conference," Information School, University of Washington and Department of Computer Science and Engineering, University of Minnesota; pp. 1-10; Nov. 6-10, 2004. cited by applicant

N. A. Lazar, "Bayesian Empirical Likelihood," Technical Report, Carnegie Mellon University, Department of Statistics, 2000; 26 pages. cited by applicant

Notess, Mark et al., "Variations2: Toward Visual Interface for Digital Music Libraries," Second International Workshop on Visual Interfaces to Digital Libraries, 2002, 6 pages. cited by applicant

O'Connor, Mary Catherine, "RFID Brings Messages to Seattle Sidewalks: An RFID system being deployed next week will send marketing and assistive information to users carrying active RFID tags," www.rfidjournal.com/article/articleview/1619/1/1, Web Page, RFID Journal (pp. 1-4). cited by applicant

P. Indyk and J. Matousek. 'Low-Distortion Embeddings of Finite Metric Spaces'. In Handbook of Discrete and Computational Geometry, pp. 177-196. CRC Press, 2004. cited by applicant

Pachet, Francois, "A Taxonomy of Musical Genres," Content-Based Multimedia Information Access Conference (RIAO), Paris, Apr. 2000, 8 pages. cited by applicant

Paek, et al., "Toward University Mobile Interaction for Shared Displays," Microsoft Research, Redmond, WA, pp. 1-4, Nov. 6-10, 2004. cited by applicant

Primary Examiner: Tsai; James T

Attorney, Agent or Firm: Blank Rome LLP

Background/Summary

PRIORITY (1) This application is entitled to the benefit of and incorporates by reference essential subject matter disclosed in U.S. application Ser. No. 16/454,668, filed on Jun. 27, 2019, which claims the benefit of U.S. Provisional 62/690,705, filed Jun. 27, 2018.

BACKGROUND

(1) This disclosure relates generally to media item playlists and more specifically to generating personalized playlists from media items from a user's social media connections.

(2) Music is an important part of interpersonal relationships, and one of the best ways to learn music is through one's interpersonal relationships. Because of this fact, some technologies display songs that have recently been played back by members of a user's social network. However, such technologies give too much importance to the influence of the social network and many of the songs that are displayed are irrelevant to the user.

SUMMARY

(3) In one embodiment, a non-transitory computer readable medium comprising computer readable

code executable by one or more processors is disclosed. The compute readable code is executable by the one or more processors to identify a user account for which a playlist is to be generated, wherein the user account is associated with a user listening history, a taste profile, and a social profile, identify one or more friend accounts linked to the user account based on the social profile, obtain an indication of a plurality of songs associated with one or more of the friend accounts, determine a listening history for the one or more plurality of songs based on the friend accounts, obtain a subset of the plurality of songs based on the listening history of the plurality of songs, and generate a playlist from the subset of the plurality of songs.

(4) In another embodiment, system for generating a playlist is disclosed. The system may include one or more processors, and one or more computer readable media comprising computer readable code executable by the one or more processors. The computer readable code is executable by the one or more processors to identify a user account for which a playlist is to be generated, wherein the user account is associated with a user listening history, a taste profile, and a social profile stored on a social media server, obtain, from a social media server, an indication of one or more friend accounts linked to the user account based on the social profile, obtain an indication of a plurality of songs associated with one or more of the friend accounts, determine a listening history for the one or more plurality of songs based on the friend accounts, obtain a subset of the plurality of songs based on the listening history of the plurality of songs, and generate a playlist from the subset of the plurality of songs.

(5) In another embodiment, method for generating a playlist is disclosed. The method includes identifying a user account for which a playlist is to be generated, wherein the user account is associated with a user listening history, a taste profile, and a social profile, identifying one or more friend accounts linked to the user account based on the social profile, obtaining an indication of a plurality of songs associated with one or more of the friend accounts, determining a listening history for the one or more plurality of songs based on the friend accounts, obtaining a subset of the plurality of songs based on the listening history of the plurality of songs, and generating a playlist from the subset of the plurality of songs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 shows, in block diagram form, a simplified network diagram according to one or more embodiments.

(2) FIG. 2 shows, in flow chart form, an example method for generating a playlist based on social data, according to one or more embodiments.

(3) FIG. 3 shows, flow chart form, an example method for building a library from which a playlist may be generated in accordance with one or more embodiments.

(4) FIG. 4 shows, in flow chart form, an example method for selecting a subset of songs based on an assigned relevance score for each song, according to one or more embodiments.

(5) FIG. 5 shows an example flow chart for managing feedback for a particular song in a playlist, according to one or more embodiments.

(6) FIG. 6 shows, in flow chart form, an example method for managing a change in relationship between a user and a friend, according to one or more embodiments.

(7) FIG. 7 shows, in flow chart form, an example method for generating a graphical user interface for the socially-driven playlist, according to one or more embodiments.

(8) FIG. 8A shows an example graphical user interface in accordance with one or more embodiments.

(9) FIG. 8B shows an additional example graphical user interface in accordance with one or more embodiments.

(10) FIG. 9 shows an example system diagram for an electronic device in accordance with one or more embodiments.

DETAILED DESCRIPTION

(11) This disclosure is directed to systems, methods, and computer readable media for providing a socially-driven playlist. In general, techniques are disclosed to provide a playlist generated from songs obtained from friend accounts linked to a user account in a social network.

(12) According to one or more embodiments, the disclosed technology addresses the need in the art to provide a customized playlist for a user that while influenced by a user account's social network, also take into account the user account's taste, by using songs or other media items from friend accounts with similar taste profiles. A user account for which a playlist is to be generated may be identified. The user account may be associated with personal data such as a user listening history, taste profile, and a social profile. One or more friend accounts may be identified as associated with the social profile, for example, in a social network. A plurality of songs are identified as being associated with the friend accounts. The plurality of songs may be refined by considering listening history of the friends. For example, only songs that have been listened to recently may be included. A playlist may then be generated from the refined plurality of songs.

(13) According to one or more embodiments, the playlist may be displayed on a graphical user interface. The graphical user interface may include a graphical depiction for one or more songs or other media items of the playlist, such as album art or a depiction of the artist. In addition, according to one or more embodiments, the graphical depiction for the one or more songs may be modified or enhanced with a graphical indication of the user or users from which the song was obtained.

(14) In the following description, for purposes of explanation, numerous specific details are set forth in order to provide a thorough understanding of the disclosed concepts. As part of this description, some of this disclosure's drawings represent structures and devices in block diagram form in order to avoid obscuring the novel aspects of the disclosed embodiments. In this context, it should be understood that references to numbered drawing elements without associated identifiers (e.g., **100**) refer to all instances of the drawing element with identifiers (e.g., **100a** and **100b**). Further, as part of this description, some of this disclosure's drawings may be provided in the form of a flow diagram. The boxes in any particular flow chart may be presented in a particular order. However, it should be understood that the particular flow of any flow diagram is used only to exemplify one embodiment. In other embodiments, any of the various components depicted in the flow chart may be deleted, or the components may be performed in a different order, or even concurrently. In addition, other embodiments may include additional steps not depicted as part of the flow chart. The language used in this disclosure has been principally selected for readability and instructional purposes, and may not have been selected to delineate or circumscribe the disclosed subject matter. Reference in this disclosure to “one embodiment” or to “an embodiment” means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment, and multiple references to “one embodiment” or to “an embodiment” should not be understood as necessarily all referring to the same embodiment or to different embodiments.

(15) It should be appreciated that in the development of any actual implementation (as in any development project), numerous decisions must be made to achieve the developers' specific goals (e.g., compliance with system and business-related constraints), and that these goals will vary from one implementation to another. It will also be appreciated that such development efforts might be complex and time consuming, but would nevertheless be a routine undertaking for those of ordinary skill in the art of image capture having the benefit of this disclosure.

(16) For purposes of this disclosure, media items are referred to as “songs.” However, in one or more embodiments, the media items referred to as “songs” could be any kind of media item, such as audio media items, video media items, visual media items, textual media items, podcasts,

interviews, radio stations, and the like.

(17) Referring to FIG. 1, a simplified block diagram is depicted of a media service **100** connected to a social media service **120** and a client device **140**, for example over a network. Client device **140** may be a multifunctional device, such as a mobile phone, tablet computer, personal digital assistant, portable music/video player, wearable device, or any other electronic device that includes a media playback system.

(18) Media service **100** may include one or more servers or other computing or storage devices on which the various modules and storage devices may be contained. Although media service **100** is depicted as comprising various components in an exemplary manner, in one or more embodiments, the various components and functionality may be distributed across multiple network devices, such as servers, network storage, and the like. Further, additional components may be used, some combination of the functionality of any of the components may be combined. Generally, media service **100** may include one or more memory devices **112**, one or more storage devices **114**, and one or more processors **116**, such as a central processing unit (CPU) or a graphical processing unit (GPU). Further processor **116** may include multiple processors of the same or different type. Memory **112** may each include one or more different types of memory, which may be used for performing device functions in conjunction with processor **116**. For example, memory **112** may include cache, ROM, and/or RAM. Memory **112** may store various programming modules during execution, including library generator module **102** and playlist generator module **104**.

(19) Media service **100** may store various personal data related to listening habits for users of the media service **100** within storage **114**. Storage **114** may include one or more physical storage devices. The physical storage devices may be located within a single location, or may be distributed across multiple locations, such as multiple servers. The personal data may include a listening history for users may be stored in a listening history store **106**. In one or more embodiments, user listening history may include information regarding songs or other media items that have been listened to or otherwise consumed, as well as timing information indicating a listening history for the songs or media items. The timing information may include, for example, a time stamp for a time at which the media item was last consumed, or may otherwise provide an indication for a specific or relative time at which the media item was last consumed. A taste profile for various users may be stored in taste profile store **108**. The taste profile contains data regarding various users' taste in media items, and may be determined based on user input, user consumption habits, user feedback, and the like. Thus, according to one or more embodiments, the taste profile may be related in part to a playback of media items. The taste profile may indicate genres of interest to the user. The taste profile may be automatically determined based on the listening history of the user. The taste profile may also be determined, at least in part, by user input. For example, a user may select particular genres that may be of interest, or may be obtained from user feedback from user listening sessions. An example is a user may up vote or down vote songs that are played in the playlist, which the system may use to ascertain trends or other information for the taste profile. In one or more embodiments, the taste profile may be provided in a full version or a compact version, such as a hash, and may be provided in a manner such that taste profiles of two users may be compared to identify a similarity between the tastes of two users. As an example, if a taste profile is presented as a hash, then in one or more embodiments, a similarity between two taste profiles may be determined by determining how many bits need to be flipped to obtain one taste profile from the other. Thus, a similarity measure between two taste profiles may accordingly be represented by the distance between two taste profiles. Some embodiments for the comparison of taste profiles are described, for example, in U.S. patent application Ser. No. 15/872,929, which is herein incorporated by reference. Additionally, or alternatively, the affinity score may be based on a comparison of the listening history of the user account and the friend account from listening history store **106**. For example, the listening history of the accounts may indicate similar genres, songs, music styles, media types, and the like, and the affinity score between the two accounts may

indicate how similar the listening history of the two accounts are.

(20) The personal data may also include a social profile for the users of the media service **100**, which may be stored in the social profile store **110**. In one or more embodiments, the social profile store may include affinity scores between various users of the media service **100**. The affinity score may indicate a closeness between two user accounts. In one or more embodiments, the affinity scores may be determined, at least in part, based on a similarity measure between the taste profiles of two users. Said another way, affinity between two users may be based in part on a similarity of the tastes of the two users, which may be calculated as described above. The social profile store **110** may also manage changes or customizations between users of the media service **100**. For example, a user may select to upgrade, downgrade, connect with a friend, or disconnect with a friend. In one or more embodiments, the social profile store **110** may manage social relationships specific to use of the media service **100**. For example, a user may have very disparate taste in music than a social connection and may prefer to disconnect from that social connection such that the music associated with that social connection is no longer used to generate playlists.

(21) Some of the data utilized for the social profile store may be obtained from a social media service **120**. The social profile may include a user account store **122** in which relationship data **124** is managed which indicates a set of user accounts linked to a particular user account, along with graphical identifiers **126** representative of each of the user accounts. For example, the friend accounts may belong to other users that the instant user has selected to be connected to in a social network. Each friend account may include their own user information, such as listening history, taste profile, and social media profile, which may also be stored in media service **100**. In one or more embodiments, the relationship data **124** may be used to supplement data in the social profile store **110**. In one or more embodiments, predetermined relationships between users indicated within relationship data **124** may supplement relationship data in social profile store **110**. As an example, a particular user's connections within social media service **120** may be used to identify and propagate social connections in social profile store **110**. Relationship data **124** may also be used to determine the affinity score between users stored in social profile store **110**. In one or more embodiments, the affinity score may be determined based on the user account social profile and the friend account social profile, such as how often the two accounts interact, a relationship level indicated by the social profiles, or the like.

(22) Returning to the media service **100**, the memory **112** includes modules that include computer readable code executable by processor **116** to cause the media service **100** to perform various tasks. As depicted, the memory **112** may include a library generator module **102** and a playlist generator module **104**. According to one or more embodiments, the library generator module **102** generates a library from which playlists may be generated for a particular user. The library may be user-specific, such as specific to a user of client device **140**. The library may be comprised of songs or other media items associated with friend accounts socially connected to the particular user account. In one or more embodiments, the songs or other media items in the library may include only a portion of all songs or media items associated with social connections to the particular user. For example, the number of songs may be limited to a particular number, and may be selected for inclusion based on a threshold relevance score. Further, the songs may be included in the library based on listening history such that most recently played songs among the friend accounts are included. According to one or more embodiments the various songs associated with friend accounts may be assigned relevance scores based on any number of factors, such as how frequently the song is played, how recently the song was played, the number of friend accounts associated with the song, an affinity score for a friend account or friend accounts associated with the song, and the like. According to one or more embodiments, selectively choosing which songs should be included in a library may optimize storage and/or processing resources required to generate customized playlists without reducing quality.

(23) Memory **112** also includes a playlist generator module **104**. In one or more embodiments, the

playlist generator module **104** may prepare customized playlists for a particular user from a library of songs generated by library generator module **102**. The playlist generator module **104** may generate a playlist on request from a user, such as a user of media player **142** of client device **140**. Additionally, or alternatively, the playlist generator module **104** may periodically prepare new playlists, such as on a daily, weekly, or monthly basis. Further, playlist generator module **104** may generate playlists in response to a trigger. Examples of a trigger include a change in a social profile of a particular user, such as the addition or deletion of a social connection, or a change in the user's taste profile. According to one or more embodiments, playlists may be generated based on a requested or determined genre. For example, the genre may be requested by a user, or may be determined based on common genres that occur within the library, or based on the user taste profile and/or the user listening history. In one or more embodiments, a playlist may be based on other seeds, such as a particular artist, tempo, style of music, decade, and the like.

(24) Upon generation of the playlist, the playlist generator module **104** may generate a graphical user interface within which to present the playlist. According to one or more embodiments, the playlist generator module **104** may generate a graphical display of the playlist such that the graphical display provides an indication of the songs of the playlist along with an indication of the friend accounts from which the songs were obtained. The graphical user interface may include a graphical depiction for one or more songs or other media items of the playlist, such as album art or a depiction of the artist. the graphical depiction for the one or more songs may be modified or enhanced with a graphical indication of the user or users from which the song was obtained. In an alternate embodiment, the graphical indication could be a graphical ribbon of users displayed above, below, or otherwise in the vicinity of the playlist in the graphical user interface. The graphical indication may be, for example, an image of the user or users. In one or more embodiments, if the song was obtained from multiple users, then a single user or multiple users may be represented by the graphical indication. In one or more embodiments, the graphical identifier of the user may be obtained from social media service **120**, such as in the graphical identifier **126** of the user account store **122**. That is, in one or more embodiments, the graphical identifier may be an image that represents the friend account in a social network.

(25) FIG. 2 shows, in flow chart form, an example method for generating a playlist based on social data, according to one or more embodiments. For purposes of explanation, the following steps will be described in the context of FIG. 1. However, it should be understood that the various actions may be taken by alternate components. In addition, the various actions may be performed in a different order. Further, some actions may be performed simultaneously, and some may not be required, or others may be added.

(26) The flow chart begins at **205** where the library generator module **102** identifies a user account associated with a user listening history, a taste profile, and a social profile. According to one or more embodiments, the user account may be a user of the media service, as shown in FIG. 1. The user account identified is a user account for which a customized playlist is to be generated. The user account may be identified, for example, in response to a request for a customized socially-driven playlist. According to one or more embodiments, the user account may be identified in response to determining that the user account is eligible for creation of a customized playlist. For example, the library generation module **102** may require that a user account have a threshold number of associated friend accounts in order to build a library and thus generate a playlist from the library. As another example, the library generation module **102** may require a threshold level of personal information stored by the music service, such as sufficient data to build a listening history and/or a taste profile for the user account.

(27) The flow chart continues at **210**, where the library generator module **102** identifies one or more friend accounts linked to the user account based on the social profile. According to one or more embodiments, the user account may be linked to friend accounts within the music service, such as a music service-specific social media network. In one or more embodiments, the user

account may additionally, or alternatively, be connected to friend accounts based on a social media network separate from the media service. In one or more embodiments, the friend accounts may be identified based on social profile store **110** and/or relationship data **124**. As will be described in greater detail below with respect to FIG. **3**, identifying the friend accounts may include identifying a subset of friend accounts socially connected to the user account, for example, based on affinity scores for the friend accounts.

(28) In one or more embodiments, other factors may be considered when identifying the friend accounts. According to one or more embodiments, contextual information may be considered. For example, the library generator module **102** may obtain geographic location information for a user account and a friend account. If a device associated with a user account and a device associated with a friend account are within a same geographic area, such as at a same social event, those friends may be selected to be used to generate the library. In one or more embodiments, the friend accounts used to generate the library may be selected automatically, or may be requested by a use, such as a user of client device **140**.

(29) At **215**, the library generator module **102** obtains an indication of a plurality of songs associated with one or more of the friend accounts. Said another way, the library generator module **102** determines a set of songs that friends of a user have been listening to. In one or more embodiments, each friend account may be associated with a plurality of songs in a number of ways. For example, a user associated with a friend account may have music or other media files stored on one or more client devices owned by the user. As another example, a user of a friend account may utilize the media service **100** for streaming songs, and may tag certain songs as being favorites, adding them to a playlist, or otherwise marking songs as being of particular interest to the user. In one example, the library generator module **102** may obtain a list of songs played by a friend account recently, such as within a predetermined time. The flow chart continues at **220**, where the library generator module **102** determines a listening history for the one or more plurality of songs based on the friend accounts. The listening history may be obtained from listening history store **106**. The listening history store **106** may be used to identify a length of time that has elapsed since the last time a friend account played a particular song, or may classify songs associated with a friend account based on a length of time since the song was last played (e.g., within the last day, within the last week, within the last month, etc.).

(30) At **225**, the library generator module **102** obtains a subset of the plurality of songs based on the listening history from friend accounts. Said another way, the library generator module **102** may build a library from the obtained subset of the plurality of songs. In one or more embodiments, the subset is obtained to produce a smaller, customized set of songs than the plurality of songs obtained at **215**. As will be described in greater detail below with respect to FIG. **4**, the listening history may be used to determine a relevance score for each song. The subset of the plurality of songs may also be obtained based on a weighted strength of the friend account from which the song was obtained.

(31) The flow chart concludes at **230** where the playlist generator module **104** generates a playlist from the subset of the plurality of songs. The playlist may be generated on demand, in response to a request from the user. Alternatively, or additionally, the playlist generator module **104** may periodically generate one or more playlists and offer the playlists to the user, such as through media player **142** of the client device **140**. Further, in one or more embodiments, playlists may be generated based on a context of the user. In one or more embodiments, contexts may be determined from sensor data from the client device **140**, or from other data associated with the user account, such as calendar entries, websites visited, email, and the like. According to one or more embodiments, context-specific customized playlists may be generated by identifying songs with tags that are associated with a particular context. geotags may be used to determine where a user is located, and songs associated with that geotag may be weighted as more relevant while the user is in the determined location. As another example, other contextual information may be used to differently weight the songs. For example, if a user is in a location such as a gym, songs with an

upbeat tempo may be weighted more heavily. By contrast, if a user is determined to be in a library, a slower tempo may be more heavily weighted. In one or more embodiments, the characteristics of the songs may be determined based on tags or other metadata stored with respect to the song. As an example, tags may identify a particular song as acoustic, aggressive, rhythmically complex, danceable, energetic, fast, happy sounding, harmonious, melodic, percussive, relaxing, smooth, and the like. In one or more embodiments, the type of song weighted heavily for a particular context may additionally be based on the user's taste profile, which may indicate a user's taste in a particular context. As an example, one user may prefer acoustic songs while driving, whereas another user may prefer aggressive songs. Some embodiments for determining relevant contexts for media items are described in U.S. patent application Ser. No. 15/720,880, which is herein incorporated by reference.

(32) Customized playlists may be generated in a number of ways. A description of some of the ways a customized playlist may be generated may be found in U.S. patent application Ser. No. 15/273,428, entitled "Personal Music Content," which is herein incorporated by reference. As will be described in greater detail below with respect to FIG. 7, the playlist generator module **104** may cause transmission of the playlist to the client device **140** along with or as part of a graphical user interface (GUI).

(33) FIG. 3 shows, flow chart form, an example method for building a library from which a playlist may be generated in accordance with one or more embodiments. Specifically, FIG. 3 shows a flow chart that depicts how the plurality of songs are obtained, for example at **215** of FIG. 2.

(34) The flow chart begins at **305** where the library generator module **102** determines a similarity measure between each friend account and the user account. In one or more embodiments, the similarity measure may be based on a comparison of a taste profile for the user account and a taste profile for the friend account. As described above, in one or more embodiments, the taste profile may be expressed in the form of a hash, where a similarity between the user account and the friend account may be determined based on how many bits must be flipped from the taste profile of the user account to achieve the taste profile of the friend account. Additionally, or alternatively, the similarity measure may be based on a comparison of the listening history of the user account and the friend account. For example, if the user account has listened to very similar music within the last week, but the user account and the friend account otherwise have disparate taste profiles, the similarity measure between the user account and the friend account may be stronger than it would be otherwise not taking into consideration the listening history of the two accounts.

(35) The flow chart continues at **310**, where the library generator module **102** determines a predetermined social relationship between each friend account and the user account. According to one or more embodiments, the predetermined social relationship may be one selected or identified by the user and/or a user associated with the friend account. For example, the social profile store **110** and/or the relationship store **124**. In one or more embodiments, a certain closeness of a relationship may indicate that songs associated with certain friend accounts may be weighted more heavily. For example, two users who interact frequently in a social network application, or two users who have self-identified as being in a relationship, or being related, may be weighted more heavily. At **315**, the library generator module **102** determines an affinity score for each friend account based on one or both of the similarity measure and the predetermined social relationship.

(36) The flow chart continues at **320**, and the library generator module **102** selects a subset of the friend accounts based on the affinity scores for each friend account. In one or more embodiments, a number of friends included in the subset may be based on a threshold affinity score. For example, a predetermined threshold for the affinity score may be used to select the subset of friends.

Alternatively, or additionally, the number of friends included in the subset may be based on a number of songs associated with the plurality of friends. For example, the friend accounts may be ranked based on affinity score, and a number of friend accounts may be selected from the top ranked friend accounts such that the number of songs associated with the selected friend accounts

constitute a threshold number of songs. The flow chart concludes at **325** when the library generator module **102** obtains the plurality of songs from the subset of friends.

(37) FIG. 4 shows, in flow chart form, an example method for selecting a subset of songs based on an assigned relevance score for each song, according to one or more embodiments. In one or more embodiments, FIG. 4 describes a more detailed view of selecting a subset of songs as in **225** of FIG. 2 for example.

(38) The flow chart begins at **405**, where a library generator module **102** determines how recently each song was played. According to one or more embodiments, songs may be considered more relevant the more recent they were played. How recently each song was played may be determined, for example, from the listening history store **106**. In one or more embodiments, the library generator module **102** may obtain an indication of how recently a song was played based on a listening history for the friend account from which the song was obtained. The listening history store may include a time stamp indicating the last time a song was played by the friend account, or may include a categorization of how recently the song was played by the friend account from which the song was obtained. The flow chart continues at **410**, and the library generator module **102** determines a number of friend accounts from which each song was obtained. In one or more embodiments, the library generator module **102** may determine the number of friends based on a listening history for each friend account from the listening history store **106**. According to one or more embodiments, the number of friend accounts may be determined out of the identified friend accounts from **210** of FIG. 2. At **415**, the library generator module **102** determines a frequency each song is played among the friend accounts. The frequency may also be determined from the listening history store **106**. According to one or more embodiments, frequency may be determined within a predetermined period of time, such as within the last day, week, or month. Frequency may be determined with respect to all friends (i.e., how frequently the song is played among all selected friend accounts, or among the combined friend accounts from which the song is obtained). Alternatively, frequency may be determined on an individual basis based on the one or more friend accounts from which the song is obtained.

(39) The flowchart continues at **420**, when the library generator module **102** assigns a relevance score for each song based on one or more of how recently the song was played, the number of friend accounts from which the song was obtained, and the frequency the song was played. The various metrics may be weighted in different manners, according to one or more embodiments. For example, how recently a song was played may have less importance than how frequently a song is played. In one or more embodiments, assigning the relevance score also includes, at **425**, weighting the relevance score based on the affinity score for the friend account(s) from which the song was obtained. For example, the affinity score between accounts of romantic partners or family members may be weighted more heavily based on the predetermined relationship.

(40) The flow chart concludes at **430**, when the library generator module **102** selects a subset of the songs based on the relevance score. According to one or more embodiments, the subset of songs may be all the songs or may be a portion of the songs for which a relevance score is determined. Further, in one or more embodiments, the subset may be determined based on a threshold number of songs to be included in the subset as well as the relevance score. That is, the songs for which a relevance score is assigned may be ranked, and a number of songs may be selected based on the threshold. The threshold may also be provided in other ways, such as a threshold data size such that the number of songs satisfies a certain data size.

(41) FIG. 5 shows an example flow chart for managing feedback for a particular song in a playlist, according to one or more embodiments. Specifically, FIG. 5 describes how relevance scores and affinity scores are impacted by feedback for a particular song from a user, once the generated playlist is presented to the user.

(42) The flow chart begins at **505** where the library generator module **102** receives feedback for a particular song of the subset of songs. The feedback may be received through a graphical user

interface, for example. According to one or more embodiments, a user may provide feedback for a particular song by providing an indication that the user enjoys the song, or that the user does not enjoy the song. In one or more embodiments, the user feedback may be more passive. For example, a user listening to the entire length of a song may be considered positive feedback, whereas a user skipping a song may be considered negative feedback.

(43) The flow chart continues at **510**, and a determination is made regarding whether the feedback is positive or negative. If the feedback is positive, then the flow chart continues at **515** and the library generator module **102** increases the affinity score for the friend account from which the particular song was obtained. The flow chart continues at **520** and the library generator module **102** increases the relevance score for the particular song. According to one or more embodiments, the feedback may indicate a relative level of positive feedback (e.g., like or love a song, rate out of five stars, and the like). In the case where the feedback includes a relative level of positive feedback, the affinity score and/or the relevance score may be increased accordingly.

(44) Returning to **510**, if the feedback is negative, then the flow chart continues at **525** and the library generator module **102** decreases the affinity score for the friend account from which the particular song was obtained. The flow chart continues at **530** and the library generator module **102** decreases the relevance score for the particular song. Similar as described above to positive feedback, in the case where the feedback includes a relative level of positive feedback, the affinity score and/or the relevance score may be increased accordingly. The flowchart concludes at **535** when the library generator module **102** augments the taste profile for the user account.

(45) FIG. 6 shows, in flow chart form, an example method for managing a change in relationship between a user and a friend, according to one or more embodiments. As described above, relationships between the user account and friend accounts may be managed by either the media service **100** or the social media service **120**, which may be separate from the media service **100** (i.e., a separate social network). According to one or more embodiments, a change in a relationship between two users may trigger a change in how media items from those users are treated for each other. Thus, a change in a relationship between two user accounts may trigger an update to the library and, thus, an update to the playlist.

(46) The flow chart begins at **605**, when the library generator module **102** receives an indication of a change in social relationship between the user account and a friend account. As an example, one or both of the user account and the friend account may indicate a change in relationship. For example, the two accounts may be newly linked. As another example, one account may select to disengage with the other or otherwise remove the friend account from a list of friend accounts. As yet another example, a type of relationship may be changed, such as a change in a romantic relationship between users of two accounts.

(47) The flowchart continues at **610** and a determination is made regarding whether the friend account is unlinked from the user account. If at **610** it is determined that the friend account is unlinked from the user account, then the flow chart continues at **615**, and the library generator module **102** removes songs obtained from the removed friend account to obtain a new plurality of songs. The flow chart continues at **620** where the library generator module **102** obtains a new subset of the new plurality of songs based on the listening history of the remaining friends.

(48) Returning to **610**, if a determination is made that the friend account is not unlinked from the user, the flow chart continues at **625**, where the library generator module **102** modifies the affinity score for the friend account from which the particular song was obtained. The flow chart continues at **630** when the library generator module **102** modifies the weight of the relevance score for songs obtained based on the modified affinity score.

(49) The flowchart concludes at **635** when the library generator module **102** generates a new playlist. According to one or more embodiments, a change in relationship may trigger a new playlist to be generated. Further, according to one or more embodiments, the new playlist may be generated the next time a playlist is periodically generated.

(50) FIG. 7 shows, in flow chart form, an example method for generating a graphical user interface (GUI) for the socially-driven playlist, according to one or more embodiments. In one or more embodiments, the GUI may improve the display of data related to a playlist by better identifying why a particular song was added to the playlist. For example, in this disclosure, the GUI may present a song or other media item along with an indication of a friend account from which the song was obtained.

(51) The flow chart begins at **705** when the playlist generator module **104** obtains a graphical representation of one or more songs from the generated playlist. The graphical representation may be obtained for each song of the playlist generated by playlist generator module **104**. According to one or more embodiments, the graphical representation of the one or more songs may include album artwork, an image of the artist, or the like. The graphical representation may be obtained from the media service **100**, or from a different location across a network. Further, a default graphical image may be utilized in the case where the graphical representation of the one or more songs are unavailable.

(52) The flow chart continues at **710** and the playlist generator module **104** obtains a graphical indication of one or more friend accounts from which the one or more songs are obtained. A graphical representation may be obtained for each friend account from which a song was obtained for the playlist. As described above, the graphical indication of the friend account may be obtained from within the media service **100**. For example, a user may select a graphical image that represents their user account. In one or more embodiments, the graphical indication may be obtained from a separate social media service **120**, such as from the graphical identifier **126** of the user account store **122**. Thus, the graphical representation of the user may be specific to the media service, or may be related to a separate social media service.

(53) Optionally, the flow chart includes, at **715**, a determination as to whether multiple friend accounts are associated with the song. If a determination is made at **715** that multiple friend accounts are not associated with the song, then the flow chart continues at **720** and the playlist generator module **104** obtains a graphical indication of the single friend account, for example, from the social media service. Returning to **715**, if a determination is made that multiple friend accounts are associated with the song, then the flow chart continues at **725**, and the playlist generator module **104** optionally selects a representative friend account. The flow chart then continues at **730** and the playlist generator module **104** generates a multi-friend indication using the representative friend account. Alternatively, if the determination is made that multiple friend accounts are associated with the song, then graphical representations for each of the friend accounts may be obtained and a multi-friend representation may be generated, for example by presenting all the graphical representations for all the friend accounts, or by showing a stacked version of the graphical representation of the friend accounts, where a first graphical representation of a first representative friend account is fully visible, and additional graphical representations of one or more additional friend accounts are partially visible. The representative friend account may be selected in a number of ways. For example, the representative friend may be selected based on a predetermined social relationship between the user account and the friend account. That is, a closeness in a social network. As another example, the representative friend may be selected based on an affinity score between the friend account and the user account (i.e., the friend account with the highest affinity score for the user account is selected). In another embodiment, the representative friend may be selected based on the listening history of the friend accounts. For example, a representative friend may be chosen based on a friend account whose listening history indicates the friend was the last user to listen to the song, the friend account which has listened to the song with the greatest frequency, and the like.

(54) The flow chart continues at **735** where the playlist generator module **104** generates a graphical user interface based on the graphical representation of the one or more songs and the graphical indication of the one or more friend accounts. According to one or more embodiments, the

graphical indication of the one or more friend accounts may be presented in association with the graphical representation of the one or more songs associated with the friend account. That is, a graphical representation of a song may be presented as associated with the graphical indication of the friend account from which it was obtained. In one or more embodiments, the graphical indication of the friend account may be presented in a badging manner, such that the graphical indication of the friend account may be overlaid on the graphical representation of the corresponding song. According to one or more embodiments, the graphical user interface may additionally, or alternatively, present the graphical representations of the friend accounts that contributed to the playlist separately from the graphical representations of the one or more songs, for example in a ribbon format. The flow chart concludes at **740** when the playlist generator module **104** causes transmission of the graphical user interface to a client device associated with the user account.

(55) FIG. **8A** shows a first example graphical user interfaces in accordance with one or more embodiments. Specifically, FIG. **8A** illustrates a multifunctional device **800** having a touch screen that displays media content, such as within media player **142** of FIG. **1**. The media player application displays content browsing user interface **802**, which may include a representation of the customized playlist generated in any manner as described above. The content browsing user interface may include a title **632** and a Friends Mix artwork **840** corresponding to the socially-driven customized playlist described herein. The content browsing user interface **802** may also include a description **824**, which may describe the playlist in more detail. According to one or more embodiments, the description **824** may also include details such as the last time the playlist was generated, a genre or category for the playlist, and the like.

(56) As described above, the content browsing user interface **802** may include the graphical representation of the song, as shown by **824A** and **824B**. Thus, the graphical representation of Song A may be Album A Art **824A**, which may be an album on which the song was released. Similarly, the graphical representation of Song B may be Album B Art **824B**, which may be an album on which the song was released. Each graphical representation of the song **824** may be presented with a graphical indication of a friend account from which the song was obtained. For purposes of the example playlist, the graphical representation of Song A **824A** may be badged by a picture of Friend A **824A**, who is associated with the friend account from which Song A was obtained for the playlist. Additionally, the graphical representation of Song B **826B** may show a multi-friend account indication **824B**, with a representative friend account indication visible (e.g., a picture of Friend B). The content browsing user interface **802** may include additional features such as a textual description of each song, as with Song A **828A** and Song B **828B**. According to one or more embodiments, the textual description may additionally include a reason the song is included in the playlist. For example, the textual description **828** may convey which of the user's friends' listening activity has resulted in the inclusion of a particular song in the playlist.

(57) The content browsing user interface **802** may also include selectable icons **830A** and **830B** which allow the user to add the song to the user's library, such as a personal library separate from the library described above which is comprised of songs obtained from friend accounts associated with the user account. The content browsing interface **802** may also include an indication of the current song playing, along with play controls, such as a pause button and a skip button, as shown by user consumption interface **840**. Thus, the user consumption interface **840** may allow a user to play and navigate through the playlist.

(58) FIG. **8B** shows a second example graphical user interfaces in accordance with one or more embodiments. Specifically, FIG. **8B** illustrates the multifunctional device **800** having a touch screen that displays media content, such as within media player **142** of FIG. **1**. The media player application displays content browsing user interface **802**, which may include a representation of the customized playlist generated in any manner as described above. In the example of FIG. **8B**, the content browsing user interface **802** additionally includes a friends ribbon, which depicts the

graphical indication of friend accounts which contributed to the playlist. Further, the content browsing user interface **802** may also include an artist ribbon **810**, which includes graphical indications of artists which contributed to the playlists. The artists may be represented, for example, by graphical representations **824**, or by different graphical representations specific to the artists rather than representative of the songs of the playlist.

(59) Referring now to FIG. **9**, a simplified functional block diagram of illustrative multifunction device **900** is shown according to one embodiment. Multifunctional device **900** may show representative components, for example, for devices of media service **100**, social media service **120**, and client device **140** of FIG. **1**. Multifunction electronic device **900** may include processor **905**, display **910**, user interface **915**, graphics hardware **920**, device sensors **925** (e.g., proximity sensor/ambient light sensor, accelerometer and/or gyroscope), microphone **930**, audio codec(s) **935**, speaker(s) **940**, communications circuitry **945**, digital image capture circuitry **950** (e.g., including camera system) video codec(s) **955** (e.g., in support of digital image capture unit), memory **960**, storage device **965**, and communications bus **970**. Multifunction electronic device **900** may be, for example, a digital camera or a personal electronic device such as a personal digital assistant (PDA), personal music player, mobile telephone, or a tablet computer.

(60) Processor **905** may execute instructions necessary to carry out or control the operation of many functions performed by device **900** (e.g., such as the generation and/or processing of images as disclosed herein). Processor **905** may, for instance, drive display **910** and receive user input from user interface **915**. User interface **915** may allow a user to interact with device **900**. For example, user interface **915** can take a variety of forms, such as a button, keypad, dial, a click wheel, keyboard, display screen and/or a touch screen. Processor **905** may also, for example, be a system-on-chip such as those found in mobile devices and include a dedicated graphics processing unit (GPU). Processor **905** may be based on reduced instruction-set computer (RISC) or complex instruction-set computer (CISC) architectures or any other suitable architecture and may include one or more processing cores. Graphics hardware **920** may be special purpose computational hardware for processing graphics and/or assisting processor **905** to process graphics information. In one embodiment, graphics hardware **920** may include a programmable GPU.

(61) Image capture circuitry **950** may include two (or more) lens assemblies **980A** and **980B**, where each lens assembly may have a separate focal length. For example, lens assembly **980A** may have a short focal length relative to the focal length of lens assembly **980B**. Each lens assembly may have a separate associated sensor element **990**. Alternatively, two or more lens assemblies may share a common sensor element. Image capture circuitry **950** may capture still and/or video images. Output from image capture circuitry **950** may be processed, at least in part, by video codec(s) **955** and/or processor **905** and/or graphics hardware **920**, and/or a dedicated image processing unit or pipeline incorporated within circuitry **965**. Images so captured may be stored in memory **960** and/or storage **965**.

(62) Sensor and camera circuitry **950** may capture still and video images that may be processed in accordance with this disclosure, at least in part, by video codec(s) **955** and/or processor **905** and/or graphics hardware **920**, and/or a dedicated image processing unit incorporated within circuitry **950**. Images so captured may be stored in memory **960** and/or storage **965**. Memory **960** may include one or more different types of media used by processor **905** and graphics hardware **920** to perform device functions. For example, memory **960** may include memory cache, read-only memory (ROM), and/or random access memory (RAM). Storage **965** may store media (e.g., audio, image and video files), computer program instructions or software, preference information, device profile information, and any other suitable data. Storage **965** may include one more non-transitory computer-readable storage mediums including, for example, magnetic disks (fixed, floppy, and removable) and tape, optical media such as CD-ROMs and digital video disks (DVDs), and semiconductor memory devices such as Electrically Programmable Read-Only Memory (EPROM), and Electrically Erasable Programmable Read-Only Memory (EEPROM). Memory **960** and

storage 965 may be used to tangibly retain computer program instructions or code organized into one or more modules and written in any desired computer programming language. When executed by, for example, processor 905 such computer program code may implement one or more of the methods described herein.

(63) The scope of the disclosed subject matter should be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.”

Claims

1. A non-transitory computer readable medium comprising computer readable code executable by one or more processors to: identify a user account for which a playlist is to be generated, wherein the user account is associated with at least one social profile of a plurality of social profiles; identify a subset of the plurality of social profiles corresponding to one or more friend accounts linked to the user account in accordance with a connection between the social profile associated with the user account and the subset of the plurality of social profiles; determine an affinity score for each of the one or more friend accounts based on a similarity in taste profile between the user account and the corresponding one or more friend accounts, and a predetermined social relationship between the user account and the corresponding one or more friend accounts, for each friend account; generate a library comprising a plurality of songs associated with the subset of the plurality of social profiles; weight a relevance score for each of the plurality of songs in the library based on the affinity score for a friend account from which each of the plurality of songs belong, and a number of one or more friend accounts associated with each of the plurality of songs; select a subset of the plurality of songs from the library based on the weighted relevance scores; and generate the playlist from the selected subset of the plurality of songs.
2. The non-transitory computer readable medium of claim 1, further comprising computer readable code to: generate a user interface for the playlist comprising a graphical representation of one or more songs of the playlist with a graphical indication of the one or more friend accounts from which the one or more songs were obtained.
3. The non-transitory computer readable medium of claim 1, further comprising computer readable code to: determine the affinity score for each friend account based on relationship data, wherein relationship data includes one or more of account interaction and relationship level indicated by the respective social profiles, wherein the plurality of songs is selected from one or more friend accounts associated with an affinity score that satisfies a predetermined threshold.
4. The non-transitory computer readable medium of claim 1, further comprising computer readable code to: rank the one or more friend accounts based on their respective affinity scores; and select a subset of friend accounts from a top of the ranked one or more friend accounts based on a number of songs associated with the one or more friend accounts constituting a threshold number of songs.
5. The non-transitory computer readable medium of claim 1, wherein the determined affinity scores for friend accounts of family members are weighted more heavily than affinity scores for friend accounts of non-family members.
6. The non-transitory computer readable medium of claim 1, further comprising computer readable code to: receive a feedback indication for a particular song of the plurality of songs from the user account; and modify the affinity score for the friend account from which the particular song was obtained based on the feedback indication.
7. A method for generating a playlist, the method, with at least one computing device, comprising: identifying a user account for which a playlist is to be generated, wherein the user account is associated with at least one social profile of a plurality of social profiles; identifying a subset of the plurality of social profiles corresponding to one or more friend accounts linked to the user account

in accordance with a connection between the social profile associated with the user account and the subset of the plurality of social profiles; determining an affinity score for each of the one or more friend accounts based on a similarity in taste profile between the user account and the corresponding one or more friend accounts, and a predetermined social relationship between the user account and the corresponding one or more friend accounts, for each friend account; generating a library comprising a plurality of songs associated with the subset of the plurality of social profiles; weighting a relevance score for each of the plurality of songs in the library based on the affinity score for a friend account from which each of the plurality of songs belong, and a number of one or more friend accounts associated with each of the plurality of songs; selecting a subset of the plurality of songs from the library based on the weighted relevance scores; and generating the playlist from the selected subset of the plurality of songs.

8. The method of claim 7, further comprising: generating a user interface for the playlist comprising a graphical representation of one or more songs of the playlist with a graphical indication of the one or more friend accounts from which the one or more songs were obtained.

9. The method of claim 7, further comprising: determining the affinity score for each friend account based on relationship data, wherein relationship data includes one or more of account interaction and relationship level indicated by the respective social profiles, wherein the plurality of songs is selected from one or more friend accounts associated with an affinity score that satisfies a predetermined threshold.

10. The method of claim 7, further comprising: ranking the one or more friend accounts based on their respective affinity scores; and selecting a subset of friend accounts from a top of the ranked one or more friend accounts based on a number of songs associated with the one or more friend accounts constituting a threshold number of songs.

11. The method of claim 7, wherein the determined affinity scores for friend accounts of family members are weighted more heavily than affinity scores for friend accounts of non-family members.

12. The method of claim 7, further comprising: receiving a feedback indication for a particular song of the plurality of songs from the user account; and modify the affinity score for the friend account from which the particular song was obtained based on the feedback indication.

13. A system for generating a playlist, comprising: one or more processors; and one or more computer readable media comprising computer readable code executable by the one or more processors to: identifying a subset of the plurality of social profiles corresponding to one or more friend accounts linked to the user account in accordance with a connection between the social profile associated with the user account and the subset of the plurality of social profiles; determining an affinity score for each of the one or more friend accounts based on a similarity in taste profile between the user account and the corresponding one or more friend accounts, and a predetermined social relationship between the user account and the corresponding one or more friend accounts, for each friend account; generating a library comprising a plurality of songs associated with the subset of the plurality of social profiles; weighting a relevance score for each of the plurality of songs in the library based on the affinity score for a friend account from which each of the plurality of songs belong, and a number of one or more friend accounts associated with each of the plurality of songs; selecting a subset of the plurality of songs from the library based on the weighted relevance scores; and generating the playlist from the selected subset of the plurality of songs.

14. The system of claim 13, further comprising computer readable code executable by the one or more processors to: generate a user interface for the playlist comprising a graphical representation of one or more songs of the playlist with a graphical indication of the one or more friend accounts from which the one or more songs were obtained.

15. The system of claim 13, further comprising computer readable code executable by the one or more processors to: determine the affinity score for each friend account based on relationship data,

wherein relationship data includes one or more of account interaction and relationship level indicated by the respective social profiles, wherein the plurality of songs is selected from one or more friend accounts associated with an affinity score that satisfies a predetermined threshold.

16. The system of claim 13, further comprising computer readable code executable by the one or more processors to: rank the one or more friend accounts based on their respective affinity scores; and select a subset of friend accounts from a top of the ranked one or more friend accounts based on a number of songs associated with the one or more friend accounts constituting a threshold number of songs.

17. The system of claim 13, further comprising computer readable code executable by the one or more processors to: receive a feedback indication for a particular song of the plurality of songs from the user account; and modify the affinity score for the friend account from which the particular song was obtained based on the feedback indication.
